

PROJECT DEMONSTRATION PHASE

ComicCraft – AI Comic Story Creator using Gemini Models

1. Introduction

The Project Demonstration Phase presents the completed ComicCraft application and explains how the system works from user input to final comic export. ComicCraft is a web-based application that generates personalized comic stories and illustrations using Gemini AI models and Stable Diffusion.

2. Demonstration Objective

The main objective of the demonstration is to show that the complete comic-generation workflow works correctly. The demonstration covers the homepage, user input, AI story generation, image generation, comic preview and PDF export.

3. Technologies Demonstrated

- FastAPI – backend routing and processing.
- HTML, CSS and Jinja2 – user interface and dynamic pages.
- Gemini Flash – structured comic panel outline generation.
- Gemini Pro – detailed narration and character dialogue generation.
- Stable Diffusion – comic-style illustration generation.
- FPDF – final comic PDF generation.

4. Demonstration Flow

The demonstration follows this sequence: user opens the homepage → enters comic details → submits the form → AI generates the comic outline and story → images are generated → comic panels are displayed → user downloads the completed comic as a PDF.

PROJECT DEMONSTRATION – HOMEPAGE & USER INPUT

1. Opening the Application

The ComicCraft application is launched locally using the FastAPI development server. The documented local address is `http://127.0.0.1:8000`. The homepage provides the starting point for creating a new AI-generated comic.

2. Homepage Demonstration

The homepage contains a clean and simple form designed to collect the information required for comic generation. The interface uses a scenic background, organized fields and clear controls.

3. User Input Fields

- **Story Prompt:** The main idea or concept of the comic.
- **Main Character Name:** The name of the hero or main character.
- **Setting:** The location of the story, such as forest, school, city or space.
- **Story Tone:** The desired mood, such as dramatic, funny, poetic or light-hearted.
- **Art Style:** The desired visual style, such as anime, pixel art, comic book or realistic.

4. Example Demonstration Input

Example story prompt: "A brave fox exploring an enchanted forest." The user can provide a character name, select a forest setting, choose a suitable tone and select an art style. After entering these details, the user submits the form to begin comic generation.

5. Backend Connection

When the form is submitted, the data is sent to the FastAPI `/generate` route using a POST request. FastAPI receives and processes the story prompt, character name, setting, tone and art style.

PROJECT DEMONSTRATION – AI COMIC GENERATION

1. Comic Outline Generation

After the user submits the form, Gemini Flash is called to create a structured five-panel comic outline. Each panel contains a panel number, title, scene description and an image-generation prompt.

2. Story and Dialogue Generation

The generated outline is then passed to Gemini Pro. Gemini Pro expands the outline into a complete comic-style story with narration and character dialogues for the panels.

3. Illustration Generation

Stable Diffusion is used to create comic-style illustrations from the image prompts generated for each panel. The generated images are saved and associated with their corresponding panels.

4. Layout Building

The layout builder organizes the generated images and story content into a structured layout. Each panel is matched with its image, panel number and text so that the comic can be displayed in the correct sequence.

5. Demonstration of Backend Workflow

The `/generate` route coordinates the major functions:

- `generate_outline()` – creates the five-panel outline.
- `generate_story()` – creates narration and dialogue.
- `generate_image()` – generates the panel illustrations.
- `build_comic_layout()` – organizes images and story content.
- `save_pdf()` – prepares the final downloadable PDF.

PROJECT DEMONSTRATION – COMIC PREVIEW

1. Comic Preview Page

After the AI generation process is completed, ComicCraft displays the generated comic on the Comic Preview Page. The panels are presented sequentially so the user can review the complete storyline.

2. Information Displayed in Each Panel

- Panel number and title.
- AI-generated comic illustration.
- Scene description.
- Caption and narration.
- Image prompt reference.

3. Review of the Generated Comic

The preview page allows the user to view the complete creative output before exporting it. The combination of images, scene descriptions, captions and narration provides the final comic experience in the browser.

4. Demonstration of Dynamic Templates

The Comic Preview Page is dynamically rendered using FastAPI's Jinja2 templates. The generated layout is passed to `comic_preview.html`, where Jinja2 loops through the panels and displays their content.

5. Expected Demonstration Result

At the end of this stage, the presenter should be able to show a complete AI-generated comic with multiple panels, corresponding illustrations, descriptions, captions and narration.

PROJECT DEMONSTRATION – PDF EXPORT & CONCLUSION

1. Download Demonstration

On the Comic Preview Page, the user can select the “Download Your Comic as PDF” option. The system compiles the comic panels, images, panel titles, scene descriptions, captions and narration into a PDF.

2. PDF Generation

The `save_pdf()` function in `exporters.py` uses the FPDF library to create the final multi-page PDF. The generated PDF is saved in the `static/exports` folder and its path is returned by the application.

3. Export Success Page

After the download is triggered, the application redirects the user to the Comic Export Success Page. This page confirms that the AI-powered comic has been successfully created and downloaded and provides an option to create another comic.

4. Final Demonstration Checklist

- Homepage loads successfully.
- User input form accepts all required comic details.
- Five-panel outline is generated.
- Narration and dialogue are generated.
- Comic illustrations are generated.
- Comic Preview displays the complete comic.
- PDF download is successfully triggered.
- Export Success page is displayed.

5. Conclusion

The Project Demonstration Phase shows the complete end-to-end functionality of ComicCraft. The system transforms a simple user prompt into a personalized comic containing structured story panels, narration, dialogue and illustrations, and finally exports the result as a downloadable PDF. This demonstrates the integration of AI models, FastAPI backend processing and a dynamic web interface.

Source: ComicCraft – AI Comic Story Creator using Gemini Models project document.